PLERIXAFOR SEACROSS 20 MG/ML SOLUTION FOR INJECTION

One ml of solution contains 20 mg plerixafor.
Each vial contains 24 mg plerixafor in 1.2 ml solution.

PHARMACEUTICAL FORM

Solution for injection.
Clear, colourless solution

CLINICAL PARTICULARS

Therapeutic indications

Adult patients

Plerixafor Seacross is indicated in combination with granulocyte-colony stimulating factor (G-CSF) to enhance mobilisation of haematopoietic stem cells to the peripheral blood for collection and subsequent autologous transplantation in adult patients with lymphoma or multiple myeloma whose cells mobilise poorly (see section Posology and method of administration).

Paediatric patients (1 to less than 18 years)

Plerixafor Seacross is indicated in combination with G-CSF to enhance mobilisation of haematopoietic stem cells to the peripheral blood for collection and subsequent autologous transplantation in children with lymphoma or solid malignant tumours, either:
- pre-emptively, when circulating stem cell count on the predicted day of collection after adequate mobilization with G-CSF (with or without chemotherapy) is expected to be insufficient with regards to desired hematopoietic stem cells yield, or
- who previously failed to collect sufficient haematopoietic stem cells (see section Posology and method of administration).

Posology and method of administration

Plerixafor Seacross therapy should be initiated and supervised by a physician experienced in oncology and/or haematology. The mobilisation and apheresis procedures should be performed in collaboration with an oncology-haematology centre with acceptable experience in this field and where the monitoring of haematopoietic progenitor cells can be correctly performed.

Age over 60 and/ or prior myelosuppressive chemotherapy and/or extensive prior chemotherapy and/or a peak circulating stem cell count of less than 20 stem cells/microliter, have been identified as predictors of poor mobilisation.

Posology

Adult

The recommended daily dose of plerixafor by subcutaneous injection (SC) is:

- 20mg fixed dose or 0.24 mg/kg of body weight for patients weighing ≤83 kg (see section Pharmacokinetic properties).
- 0.24 mg/kg of body weight for patients weighing >83 kg.

Paediatric (1 to less than 18 years)

The recommended daily dose of plerixafor by subcutaneous injection (SC) is:

- 0.24 mg/kg of body weight (see section Pharmacodynamic properties)

Each vial of plerixafor is filled to deliver 1.2 ml of 20 mg/ml plerixafor aqueous solution for injectioncontaining 24 mg of plerixafor.
Plerixafor has to be drawn up into a syringe size type which should be selected according to the weight of the patient.
For low weight patients, up to 45 kg of body weight, 1 ml syringes for use in infant patients can be used.
This type of syringe has major graduations for 0.1 ml and minor graduations for 0.01 ml and therefore is suitable to administer plerixafor, at a dose of 240 µg/kg, to paediatric patients of at least 9 kg body weight.

For patients of more than 45 kg, a 1ml or 2ml syringe with graduations that allow a volume to 0.1 ml to be measured can be used.

It should be administered by subcutaneous injection 6 to 11 hours prior to initiation of each apheresis following 4-day pre-treatment with G-CSF. Plerixafor Seacross has been commonly used for 2 to 4 (and up to 7) consecutive days.

The weight used to calculate the dose of plerixafor should be obtained within 1 week before the first dose of plerixafor. The dose of plerixafor has been calculated based on body weight in patients up to 175% of ideal body weight. Plerixafor dose and treatment of patients weighing more than 175% of ideal body weight have not been investigated. Ideal body weight can be determined using the following equations:

male (kg): 50 + 2.3 x ((Height (cm) x 0.394) – 60);
female (kg): 45.5 + 2.3 x ((Height (cm) x 0.394) – 60).

Based on increasing exposure with increasing body weight, the plerixafor dose should not exceed 40 mg/day.

Recommended concomitant medicinal products
All patients received daily morning doses of 10 μg/kg G-CSF for 4 consecutive days prior to the first dose of plerixafor and on each morning prior to apheresis.

Special populations
Renal impairment
Patients with creatinine clearance 20 - 50 ml/min should have their dose of plerixafor reduced by one-third to 0.16 mg/kg/day (see section Pharmacokinetics properties). Clinical data with this dose adjustment are limited. There is insufficient clinical experience to make alternative posology recommendations for patients with a creatinine clearance < 20 ml/min, as well as to make posology recommendations for patients on haemodialysis.

Based on increasing exposure with increasing body weight the dose should not exceed 27 mg/day if the creatinine clearance is lower than 50 ml/min.

Paediatric population
The safety and efficacy of Plerixafor Seacross in children (1 to less than 18 years) were studied (see section Undesirable effects and Pharmacokinetic properties).

Elderly patients (> 65 years old)
No dose modifications are necessary in elderly patients with normal renal function. Dose adjustment in elderly patients with creatinine clearance ≤ 50 ml/min is recommended (see Renal impairment above). In general, care should be taken in dose selection for elderly patients due to the greater frequency of decreased renal function with advanced age.

Method of administration
Plerixafor Seacross is for subcutaneous injection. Each vial is intended for single use only.

Vials should be inspected visually prior to administration and not used if there is particulate matter or discolouration. Since Plerixafor Seacross is supplied as a sterile, preservative-free formulation, aseptic technique should be followed when transferring the contents of the vial to a suitable syringe for subcutaneous administration.

Contraindications
Hypersensitivity to the active substance or to any of the excipients listed.

Special warnings and precautions for use
Tumour cell mobilisation in patients with lymphoma and multiple myeloma
When Plerixafor Seacross is used in conjunction with G-CSF for haematopoietic stem cell mobilisation in patients with lymphoma or multiple myeloma, tumour cells may be released from the marrow and subsequently collected in

the leukapheresis product. Results showed that, in case tumour cells are mobilised, the number of tumour cells mobilised is not increased upon Plerixafor Seacross plus G-CSF compared to G-CSF alone.

Tumour cell mobilisation in leukaemia patients
In a compassionate use programme, Plerixafor Seacross and G-CSF have been administered to patients with acute myelogenous leukaemia and plasma cell leukaemia. In some instances, these patients experienced an increase in the number of circulating leukaemia cells. For the purpose of haematopoietic stem cell mobilisation, plerixafor may cause mobilisation of leukaemic cells and subsequent contamination of the apheresis product. Therefore, plerixafor is not recommended for haematopoietic stem cell mobilisation and harvest in patients with leukaemia.

Haematological effects
Hyperleukocytosis
Administration of Plerixafor Seacross in conjunction with G-CSF increases circulating leukocytes as well as haematopoietic stem cell populations. White blood cell counts should be monitored during Plerixafor Seacross therapy.
Clinical judgment should be exercised when administering Plerixafor Seacross to patients with peripheral blood neutrophil counts above 50 x 10^9/L.

Thrombocytopenia
Thrombocytopenia is a known complication of apheresis and has been observed in patients receiving Plerixafor Seacross. Platelet counts should be monitored in all patients receiving Plerixafor Seacross and undergoing apheresis.

Allergic reactions
Plerixafor Seacross has been uncommonly associated with potential systemic reactions related to subcutaneous injection such as urticaria, periorbital swelling, dyspnoea, or hypoxia (see section Undesirable effects). Symptoms responded to treatments (e.g., antihistamines, corticosteroids, hydration or supplemental oxygen) or resolved spontaneously. Cases of anaphylactic reactions, including anaphylactic shock, have been reported from world-wide post-marketing experience. Appropriate precautions should be taken because of the potential for these reactions.

Vasovagal reactions
Vasovagal reactions, orthostatic hypotension, and/or syncope can occur following subcutaneous injections (see section Undesirable effects). Appropriate precautions should be taken because of the potential for these reactions.

Effect on the spleen
The effect of plerixafor on spleen size in patients has not been specifically evaluated. Cases of splenic enlargement and/or rupture have been reported following the administration of Plerixafor Seacross in conjunction with growth factor G-CSF. Individuals receiving Plerixafor Seacross in conjunction with G-CSF who report left upper abdominal pain and/or scapular or shoulder pain should be evaluated for splenic integrity.

Sodium
Plerixafor Seacross contains less than 1 mmol sodium (23 mg) per dose, i.e. essentially ‘sodium- free’.

Interaction with other medicinal products and other forms of interaction
No interaction studies have been performed. Plerixafor was not metabolised by P450 CYP enzymes, did not inhibit or induce P450 CYP enzymes. Plerixafor did not act as a substrate or inhibitor of P-glycoprotein.

In patients with non-Hodgkin’s lymphoma, the addition of rituximab to a mobilization regimen of plerixafor and G-CSF did not impact patient safety or CD34+ cell yield.

Fertility, pregnancy and lactation
Women of childbearing potential
Women of childbearing potential have to use effective contraception during treatment.

Pregnancy
There are no adequate data on the use of plerixafor in pregnant women.
Based on the pharmacodynamic mechanism of action, plerixafor is suggested to cause congenital malformations when administered during pregnancy.
Studies in animals have shown teratogenicity. Plerixafor Seacross should not be used during pregnancy unless the clinical condition of the woman requires treatment with plerixafor.

Breastfeeding
It is not known whether plerixafor is excreted in human milk. A risk to the suckling child cannot be excluded. Breast-feeding should be discontinued during treatment with Plerixafor Seacross.

Fertility
The effects of plerixafor on male and female fertility are not known.

Effects on ability to drive and use machines
Plerixafor Seacross may influence the ability to drive and use machines. Some patients have experienced dizziness, fatigue or vasovagal reactions; therefore caution is advised when driving or operating machines.

Undesirable effects
Tabulated list of adverse reactions
Adverse reactions are listed by System Organ Class and frequency. Frequencies are defined according to the following convention: very common; common; uncommon; rare; very rare; not known.

Blood and lymphatic system disorders	
Not known	Splenomegaly, splenic rupture (see section Special warnings and precautions for use) **
Immune system disorders	
uncommon	Allergic reaction* Anaphylactic reactions, including anaphylactic shock (see section Special warnings and precautions for use) **
Psychiatric disorders	
Common	insomnia
Uncommon	Abnormal dreams, nightmares
Nervous system disorders	
common	Dizziness, headache
Gastrointestinal disorders	
Very common	Diarrhea, nausea
common	Vomiting, abdominal pain, stomach discomfort, dyspepsia, abdominal distention, constipation, flatulence, hypoaesthesia oral, dry mouth
Skin and subcutaneous tissue disorders	
Common	Hyperhidrosis, erythema
Musculoskeletal and connective tissue disorders	
Common	Arthralgia, musculoskeletal pain
General disorders and administration site conditions	
Very common	Injection and infusion site reactions
common	Fatigue, malaise

** From post-marketing experience

Description of selected adverse reactions
Myocardial infarction
Oncology patients experienced myocardial infarctions after haematopoietic stemcell mobilisation with plerixafor and G-CSF. All events occurred at least 14 days after last Plerixafor Seacross administration. Additionally, patients in the compassionate use programme experienced myocardial infarction following haematopoietic stem cell mobilisation with plerixafor and GCSF. These events occurred 4 days after last Plerixafor Seacross administration.

Lack of temporal relationship in patients coupled with the risk profile of patients with myocardial infarction does not suggest Plerixafor Seacross confers an independent risk for myocardial infarction in patients who also receive G-CSF.

Hyperleukocytosis
White blood cell counts of 100 x 10^9/l or greater were observed, on the day prior to or any day of apheresis, in patients receiving Plerixafor Seacross. No complications or clinical symptoms of leukostasis were observed.

Vasovagal reactions
Less than 1% of patients experienced vasovagal reactions (orthostatic hypotension and/or syncope) following subcutaneous administration of plerixafor doses ≤ 0.24 mg/kg. The majority of these events occurred within 1 hour of Plerixafor Seacross administration.

Gastrointestinal disorders
There have been rare reports of severe gastrointestinal events, including diarrhoea, nausea, vomiting, and abdominal pain.

Paraesthesia
Paraesthesia is commonly observed in oncology patients undergoing autologous transplantation following multiple disease interventions.

Elderly patients
No notable differences in the incidence of adverse reactions were observed in these elderly patients when compared with younger ones.

Paediatric population
The safety profile in this paediatric was consistent with what has been observed in adults.

Overdose
No case of overdose has been reported. Based on limited data at doses above the recommended dose and up to 0.48mg/kg the frequency of gastrointestinal disorders, vasovagal reactions, orthostatic hypotension, and/or syncope may be higher.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Other immunostimulants; ATC code: L03AX16
<u>Mechanism of action</u>
Plerixafor is a bicyclam derivative, a selective reversible antagonist of the CXCR4 chemokine receptor and blocks binding of its cognate ligand, stromal cell-derived factor-1α (SDF-1α), also known as CXCL12.
Plerixafor-induced leukocytosis and elevations in circulating haematopoietic progenitor cell levels are thought to result from a disruption of CXCR4 binding to its cognate ligand, resulting in the appearance of both mature and pluripotent cells in the systemic circulation. CD34+ cells mobilised by plerixafor are functional and capable of engraftment with long-term repopulating capacity.

Pharmacokinetic properties

<u>Absorption</u>
Plerixafor is rapidly absorbed following subcutaneous injection, reaching peak concentrations in approximately 30 - 60 minutes (tmax). Following subcutaneous administration of a 0.24 mg/kg dose to patients after receiving 4-days of G-CSF pre-treatment, the maximal plasma concentration (Cmax) and systemic exposure (AUC0-24) of plerixafor were 887 ± 217 ng/ml and 4337 ± 922 ng.hr/ml, respectively.

<u>Distribution</u>
Plerixafor is moderately bound to human plasma proteins up to 58%. The apparent volume of distribution of plerixafor in humans is 0.3 l/kg demonstrating that plerixafor is largely confined to, but not limited to, the extravascular fluid space.

Biotransformation

Plerixafor is not metabolized in vitro using human liver microsomes or human primary hepatocytes and does not exhibit inhibitory activity in vitro towards the major drug-metabolising CYP450 enzymes (1A2, 2A6, 2B6, 2C8, 2C9, 2C19, 2D6, 2E1, and 3A4/5). In in vitro studies with human hepatocytes, plerixafor does not induce CYP1A2, CYP2B6, and CYP3A4 enzymes. These findings suggest that plerixafor has a low potential for involvement in P450-dependent drug-drug interactions.

Elimination

The major route of elimination of plerixafor is urinary. Following a 0.24 mg/kg dose in healthy individuals with normal renal function, approximately 70% of the dose was excreted unchanged in urine during the first 24 hours following administration. The elimination half-life (t1/2) in plasma is 3 - 5 hours. Plerixafor did not act as a substrate or inhibitor of P-glycoprotein in an in vitro study with MDCKII and MDCKII-MDR1 cell models.

Special populations

Renal impairment

Following a single dose of 0.24 mg/kg plerixafor, clearance was reduced in patients with varying degrees of renal impairment and was positively correlated with creatinine clearance (CrCl). Mean values of AUC_{0-24} of plerixafor in subjects with mild (CrCl 51 - 80 ml/min), moderate (CrCl 31 - 50 ml/min) and severe (CrCl ≤ 30 ml/min) renal impairment were 5410, 6780, and 6990 ng.hr/ml, respectively, which were higher than the exposure observed in healthy patients with normal renal function (5070 ng.hr/ml). Renal impairment had no effect on Cmax.

Gender

A population pharmacokinetic analysis showed no effect of gender on pharmacokinetics of plerixafor.

Elderly

A population pharmacokinetic analysis showed no effect of age on pharmacokinetics of plerixafor.

Paediatric population

The pharmacokinetics of plerixafor were evaluated in paediatric patients (1 to less than 18 years) with solid tumours at subcutaneous doses of 0.16, 0.24 and 0.32 mg/kg with standard mobilisation (G-CSF plus or minus chemotherapy). Based on population pharmacokinetic modeling and similar to adults, µg/kg-based dosage results in increase in plerixafor exposure with increasing body weight in paediatric patients. At the same weight-based dosing regimen of 240 µg/kg, the plerixafor mean exposure ($_{AUC0-24h}$) is lower in paediatric patients aged 2 to <6 years (1410 ng.h/mL), 6 to <12 years (2318 ng.h/mL), and 12 to <18 years (2981 ng.h/mL) than in adults (4337 ng.h/mL). Based on population pharmacokinetic modeling, the plerixafor mean exposures (AUC_{0-24h}) in paediatric patients aged 2 to <6 years (1905 ng.h/mL), 6 to <12 years (3063 ng.h/mL), and 12 to <18 years (4015 ng.h/mL), at the dose of 320 µg/kg are closer to the exposure in adults receiving 240 µg/kg.
However, mobilization of PB CD34+ count was observed in stage 2 of the trial.

PHARMACEUTICAL PARTICULARS

Incompatibilities

In the absence of compatibility studies, Plerixafor Seacross must not be mixed with other medicinal products.

Special precautions for storage

Store below 30°C

After opening

From a microbiological point of view the product should be used immediately.
If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

Nature and contents of container

Clear type I glass 2 ml vial with a 13mm Teflon coated chlorobutyl rubber stopper and aluminium seal with a 13mm plastic blue flip-off cap. Each vial contains 1.2 ml solution.
Pack size of 1 vial packed in colorful, carton folded box.

Special precautions for disposal
Any unused product or waste material should be disposed of in accordance with local requirements.

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